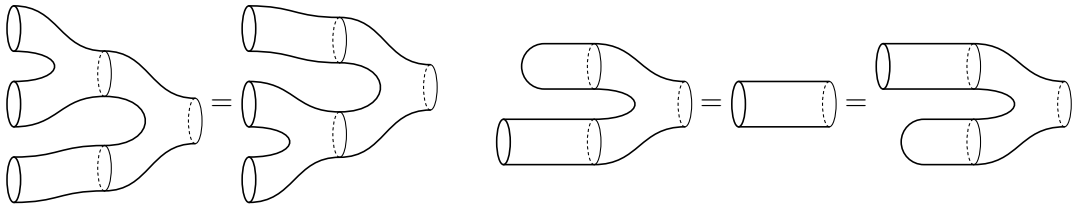
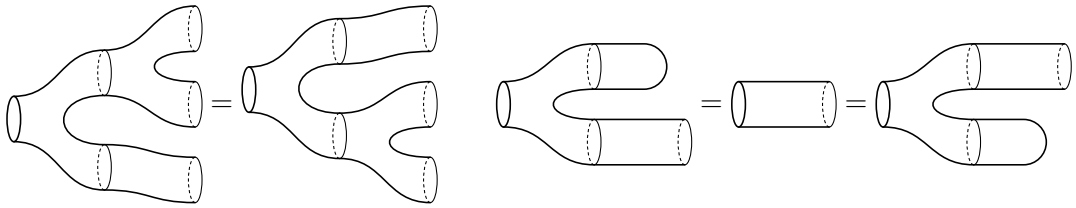
2d TQFTs and commutative Frobenius algebras

The relations of **2Cob**, written with `plumber`. Every equation is a chain of `\PL` layers; the last column shows what `reduce` makes of the left-hand side.

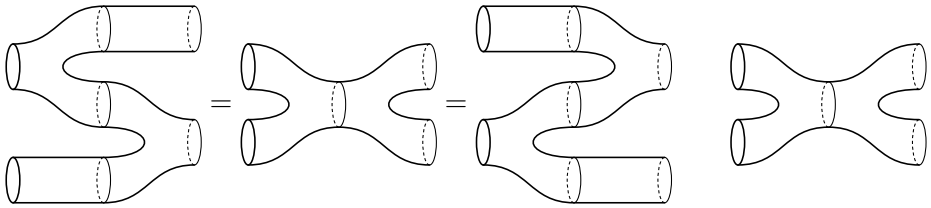
Associativity and unit



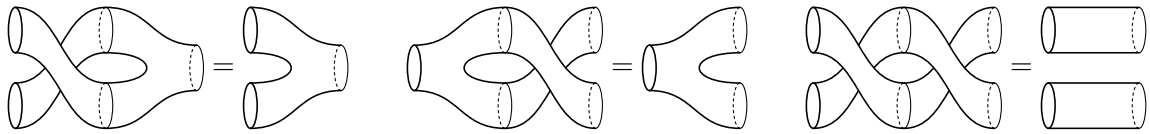
Coassociativity and counit



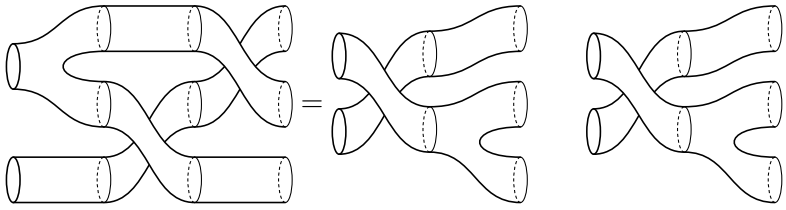
Frobenius relation



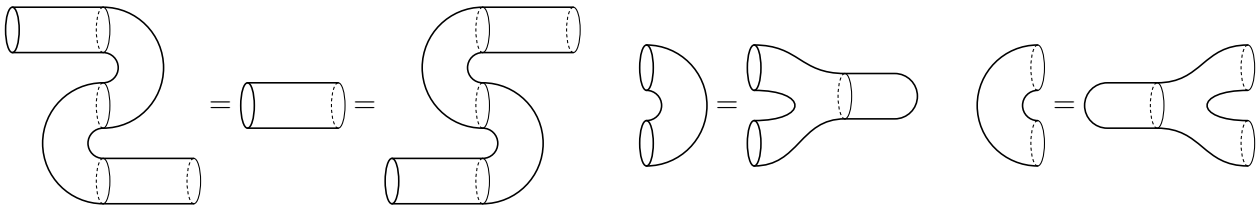
Commutativity and cocommutativity



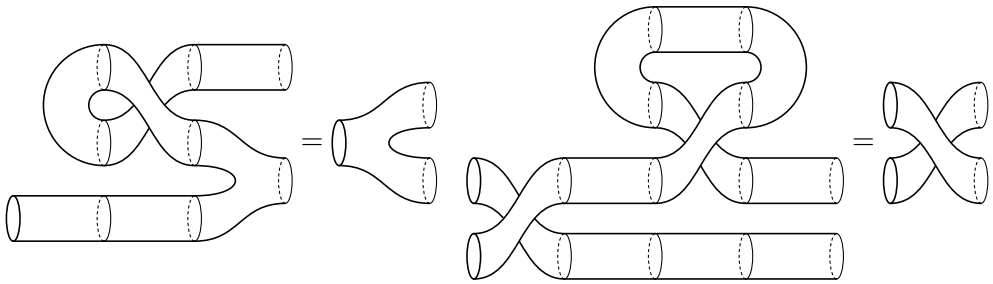
Naturality of the swap



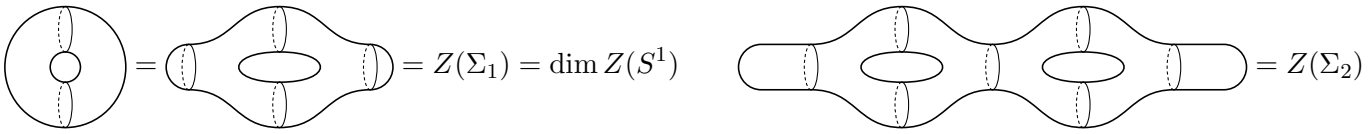
Duality: the snake (zig-zag) identities



Composites with crossings

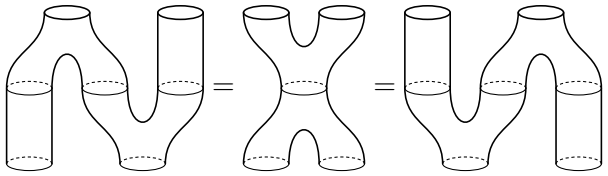


Closed surfaces

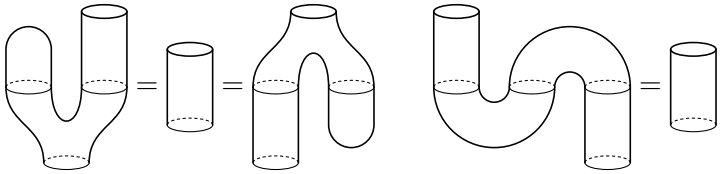


The same, read from top to bottom

Frobenius relation



Unit, counit and snake identities



A genus 2 surface with three inputs and two outputs, and its normal form

